

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

30th November 2012

Subject ST8 - General Insurance: Pricing

Time allowed: Three hours (14.45* – 18.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception*
2. ** You have 15 minutes at the start of the examination in which to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor*
4. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)**
- a)** Define an insured peril. (1)
 - b)** Define rating factor. How does it relate to risk factor? (1)
 - c)** List perils and rating factors commonly used in motor property damage cover. (5)
[7]
- Q. 2)**
- a)** In a country, motor insurance is sold with premiums fixed at outset of each policy and with renewal premiums fixed on the basis of past years' experience.
A company that sells most lines with a majority of sales in motor is working on a pilot project for redesigning the pricing of motor property damage cover. For each vehicle, the project requires installation of a telecommunication device which will communicate on real-time basis with a remote satellite. The device will require very little power to send signals and will continue to function even when the vehicle is not in use. The device will track the following items of information for each vehicle:
 - Location of vehicle
 - Distance travelled
 - Time of day
 - Speed of driving

The device will provide records every minute for each vehicle in respect of the above items.

What benefits could be derived by insurers by installing such devices in insured vehicles? (4)
 - b)** The company would like to make use of the information gathered over 6 months from 200 vehicles to analyse the appropriateness of premiums being charged currently.

What could be specific challenges in relation to data in such a study? (2)
[6]
- Q. 3)** Discuss the pros and cons to a customer of buying several covers together in a package policy. (4)
[4]
- Q. 4)** Describe with examples capital market products available to insurers so they are able to transfer risk to capital markets rather than insurance markets. (6)
[6]

- Q. 5)** **a)** What are soft premium rates in context of insurance cycle? (1)
- b)** What are likely outcomes while a line of business has soft rates, before it moves towards harder rates, in terms of
- Risk segmentation
 - Demand for insurance
 - Loss ratios
 - Any premium deficiency reserves
 - Capital required
 - Reinsurance support
 - Number of insurers offering cover
- (7)
- [8]**
- Q. 6)** **a)** Define a latent claim. (1)
- b)** Describe the claim characteristics in respect of latent claims. (7)
- [8]**
- Q. 7)** An actuarial analyst in an insurance company is working on claims data to identify trends in data which could be used in the pricing exercise. State likely sources of distortions that should be allowed for in the data while drawing any conclusions for pricing.
- [3]**
- Q. 8)** An insurer specialises in travel policies. It has now started to market a product which indemnifies, subject to a cap, against unplanned health treatment costs incurred while travelling abroad.
- a)** State risks specific to this new product the insurer is likely to take into account while pricing. (2)
- b)** State what can be done to mitigate the risks. (2)
- [4]**
- Q. 9)** **a)** Define underinsurance. (1)
- b)** How does the principle of average help insurers to settle claims on underinsured covers? (2)
- [3]**

- Q. 10)** You are an assistance pricing actuary of a reinsurance company. The pricing actuary has given you an assignment to derive the expected claim cost for motor third party liability reinsurance of an Indian insurer under a number of options. He has also provided a statistics using the past experience of the insurance company for the portfolio to be reinsured. You may assume that all required adjustments have been done to the experience data. The statistics is the total expected aggregate claim for the insurer in the coverage period for various aggregate limits with excess 1,000,000 as following:

Aggregate Limit with excess 1,000,000	Expected Aggregate Claim Cost
1,000,000	700,000
2,000,000	900,000
3,000,000	1,050,000
4,000,000	1,090,000
Unlimited	1,110,000

Derive the expected claim cost for the reinsurance layer of Rs 1,000,000 Xs 1,000,000 with following options:

- i) Unlimited free reinstatements with 1,000,000 Annual Aggregate Deductible (1)
 - ii) 1 free reinstatement (1)
 - iii) 1 free reinstatement with 1,000,000 Annual Aggregate Deductible (2)
- [4]**

- Q. 11)** You are a pricing actuary in a motor insurance company. One of your colleagues in senior management commented the following, “In my view motor pricing is a very simple exercise. Collect the data, correct the errors in the data, feed the data in statistical analysis software to run GLM and get the pricing estimates.”

Outline the points which you should mention in your reply to your colleague to tell him about

- i) The steps which need to be done to prepare the data for analysis and the initial analysis which need to be performed before the GLM analysis is done (6)
- ii) The considerations and adjustments to the GLM analysis for pure claim cost estimation (8)
- iii) The further considerations in using the results from the GLM exercise to calculate the final premium for the motor portfolio (6)

[20]

- Q. 12)** Outline the advantages and disadvantages of using the frequency-severity approach for pricing compared to the burning cost approach.

[3]

Q. 13) Let N be the random variable for number of claims, S be the random variable for severity and $PP = NS$ be the random variable for Pure Premium. We want to derive the standard for full credibility so that the mean of the random variable lies within $k\%$ of the observed mean with probability P .

i) Show that the standard for full credibility for N is given by $n_0 = \left(\frac{y}{k}\right)^2$ where

$$y = \Phi^{-1}\left(\frac{1+P}{2}\right) \text{ if } N \text{ follows Poisson} \quad (3)$$

ii) Show that the standard for full credibility for S is given by $n_0 CV_S^2$ where CV_S is the Coefficient of variation for S (3)

iii) Show that the standard for full credibility for PP is given by $n_0(1+CV_S^2)$ where CV_S is the Coefficient of variation for S (4)

iv) Derive the standard for full credibility for N , S and FF where $P = 95\%$, $k = 5\%$ and S follows Gamma (5, 10000) with mean 50000. (2)
[12]

Q. 14) Question

i) Prove that the exposure curve for a random variable X , $G(x)$ follows the equation

$$G'(x) = \left(\frac{1-F(x)}{E(X)} \right), \text{ where } F(x) \text{ is the CDF of } X \text{ and } E(X) \text{ is the expected value of } X. \quad (3)$$

You are the pricing actuary for a reinsurance company. An insurance company has submitted the following exposure profile for personal homeowners portfolio:

Sum Insured			
Lower Limit	Upper Limit	Original Premium	Original Loss Ratio
0	100	10,000	60%
100	150	3,000	60%
150	200	2,000	60%
200	250	1,000	60%
250	300	500	60%
300	400	600	60%
400	500	400	60%
500	1,000	1,000	60%

The exposure is based on a portfolio at 30th June 2012. The reinsurance coverage period is 1st January 2013 to 31st December on Loss Occurrence Basis. The annual inflation between 2012 and 2013 is 10%.

- ii) Comment if the exposure curve for a 2012 portfolio is suitable to use in 2013. (1)
- iii) Calculate the pure claim cost based on the above exposure profile where the exposure curve $G(x) = \left(\frac{1-0.1^x}{1-0.1} \right)$ for $0 \leq x \leq 1$, for the layer 250 Xs 50. Using mid-point of the sum insured bands as an approximation for the sum insured for all risks in a band. (4)
- iv) Comment on the possible problem in using mid-point of the sum insured bands as an approximation for the sum insured for all risks in a band. (2)
- v) Outline possible approaches to overcome problem. (2)
- [12]**
